

May 23, 2026

1. The question

Market structure (reform calendar in one paragraph).

Who actually sets the price (NOT a priori CCGT).

Why this matters.

Outcomes we care about, and why.

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2. Variation across hours: where granularity has economic content

Intuition.

Classifier.

What we see.

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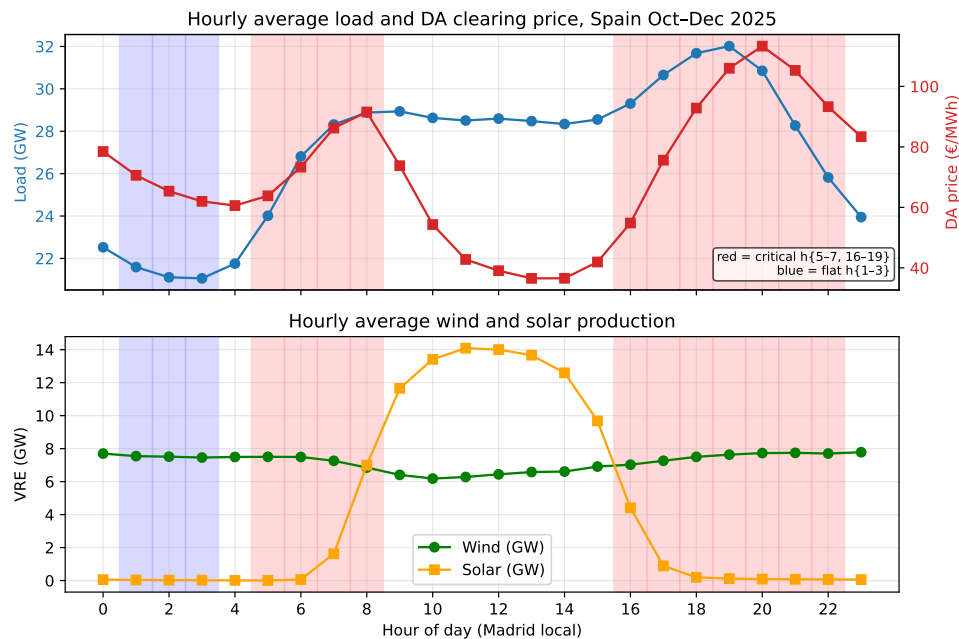


Figure 1:

Validation.

3. Variation across firms and technologies: how do CCGTs bid?

The object.

Metric: D_w .

Decomposition: Δp vs Δq .

What we see (post-MTU15-DA, critical hours, CCGT, Oct–Dec 2025).

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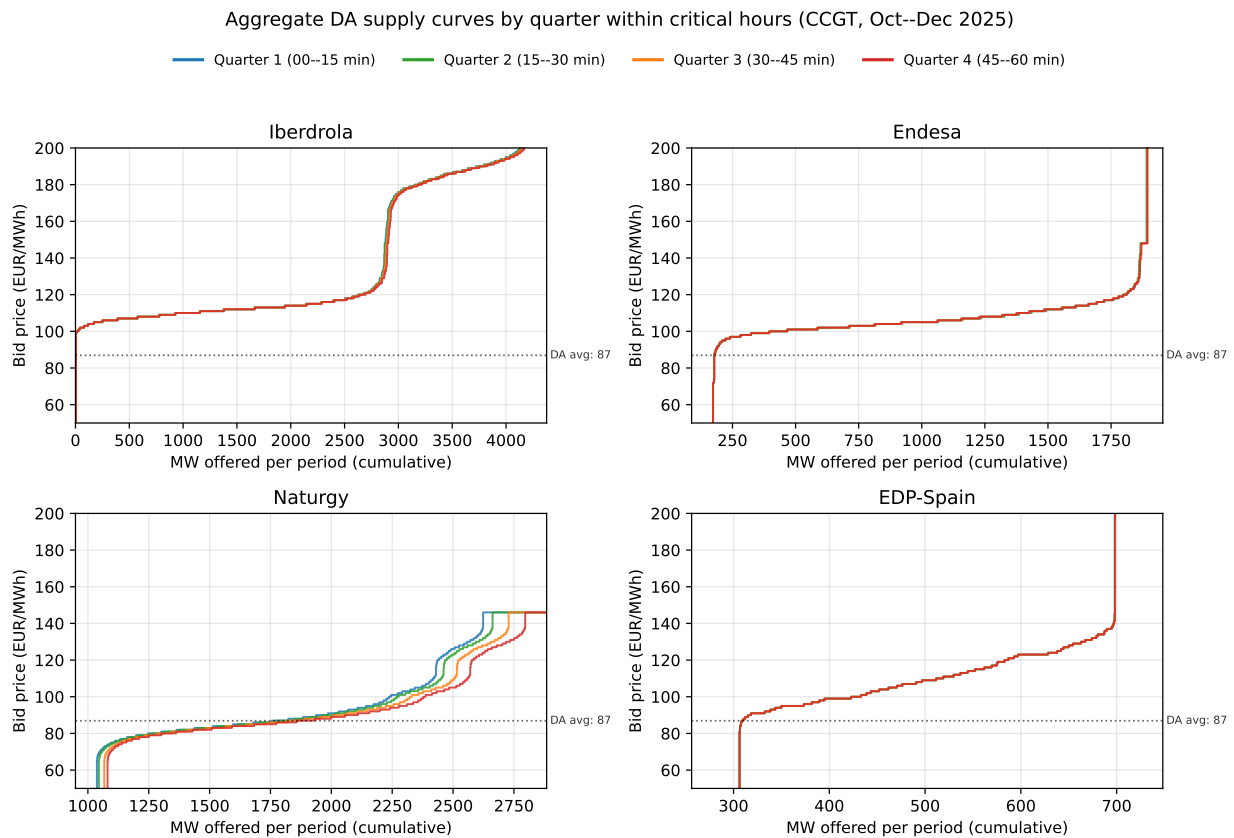


Figure 2:

Operational vs strategic decomposition.

4. The 2025 regulatory shock: blackout and reinforced operation

The blackout (28 April 2025).

What “reforzada” does in practice.

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What this lets us claim.

5. Reform windows we use, and outcomes we measure in each

Strategy in one sentence.

The reform calendar and our windows.

Reform	Date	Window and why

Outcomes we measure in each window.

Outcome	What it tells us / why it matters

6. MTU15 effect: critical/flat identification (ID15 and DA15)

Strategy. (Gap to write: 3–4 sentences recapping the critical/flat identification — critical hours $\{5, 6, 7, 8, 16, \dots, 22\}$ are where within-hour residual demand varies and MTU15 has economic content; flat hours $\{1, 2, 3\}$ are the within-day control where MTU15 should not matter; the (critical – flat) differential across the reform date isolates the MTU15 effect from any day-level shock that hits both classes proportionally.)

6.1 Specifications

Let u index a unit, d a date, p a market period (hour pre-MTU15, quarter post), h the clock-hour, s the IDA session, T_r the reform date, $\mathcal{C}$ the critical-hour set. All sell-side, in-band $|p_{\text{bid}} - \text{MCP}| \leq 140$ EUR/MWh.

Spec A — per-curve functional outcomes $Y \in \{\sigma_p, N_{\text{eff}}\}$, panel $(u, d, p[, s])$:

$$Y_{u,d,p[,s]} = \alpha + \alpha_u + \delta_d + \gamma_h + \beta \mathbf{1}\{d \geq T_r\} + \theta \mathbf{1}\{d \geq T_r\} \cdot \mathbf{1}\{h \in \mathcal{C}\} + \varepsilon$$

α_u unit FE absorbed by within-unit demeaning; γ_h absorbs the crit-flat baseline; SEs clustered by date. θ is the DiD.

Spec B — aggregate clearing-price DiD, panel $(d, p[, s])$ over the OMIE clearing price P :

$$P_{d,p[,s]} = \alpha + \gamma_h + \beta \mathbf{1}\{d \geq T_r\} + \theta \mathbf{1}\{d \geq T_r\} \cdot \mathbf{1}\{h \in \mathcal{C}\} + \varepsilon$$

Spec C — within-hour dispersion (post-only; pre has one period per hour, SD undefined). For each (unit, date, clock-hour [, session]) with all 4 quarters present:

$$D_{\text{price}}^{(u,d,h)} = \text{sd}_q\{\bar{p}_{u,d,h,q}\}, \quad D_{\text{qty}}^{(u,d,h)} = \frac{\text{sd}_q\{m_{u,d,h,q}\}}{\text{mean}_q\{m_{u,d,h,q}\}},$$

where $\bar{p}_{u,d,h,q}$ is the unit's MW-weighted in-band mean price in quarter q and $m_{u,d,h,q}$ is its in-band MW. Regress on the critical indicator:

$$D^{(u,d,h)} = \alpha + \alpha_u + \delta_d + \theta \mathbf{1}\{h \in \mathcal{C}\} + \varepsilon$$

System-level analog drops the unit dimension and uses the clearing-price SD across the 4 quarters.

6.2 Sample windows (reforzada handled by tight pre/post)

Reform	Pre window	Post window	Reforzada
MTU15-IDA	2024-12-19 → 2025-03-18 (3 mo, ISP15)	2025-03-19 → 2025-04-27 (40 d, clean)	absent both arms
MTU15-DA	2025-07-01 → 2025-09-30 (3 mo)	2025-10-01 → 2025-12-31 (3 mo)	active both arms, level constant

For MTU15-IDA the post-window is pre-blackout (reforzada absent throughout) so θ identifies the MTU15-IDA effect cleanly. For MTU15-DA both arms sit under continuous reforzada (active since 2025-04-28); the DiD critical-flat interaction strips the level shift as long as the reforzada critical-vs-flat differential is stable across the cutover.

6.3 Spec A — per-curve σ_p and N_{eff}

	(1) Price spread σ_p	(2)	(3) Tranche count N_{eff}	(4)
	MTU15-IDA	MTU15-DA	MTU15-IDA	MTU15-DA
<i>Panel A: θ by technology (firm-pooled)</i>				
CCGT	2.45*** (0.58)	1.30* (0.75)	0.79*** (0.30)	0.44*** (0.06)
Hydro	1.17*** (0.44)	1.69*** (0.29)	0.21*** (0.06)	−0.04 (0.04)
Hydro pump	0.25 (0.28)	−0.63* (0.36)	0.18** (0.08)	−0.05 (0.03)
(Pooled)	1.71*** (0.34)	1.36*** (0.34)	0.10 (0.12)	0.14*** (0.04)
<i>Panel B: θ by CCGT firm (Big-4)</i>				
IB	6.46*** (1.72)	3.01*** (1.12)	0.15** (0.06)	0.32*** (0.10)
GE	2.10** (0.87)	−1.47*** (0.32)	0.15*** (0.04)	−0.03*** (0.01)
GN	2.88*** (0.67)	35.00*** (2.26)	1.84*** (0.50)	3.13*** (0.23)
HC	−1.03 (1.60)	−12.96*** (1.40)	−0.15 (0.14)	−0.27*** (0.05)
Unit FE	Yes	Yes	Yes	Yes
Date FE	No	No	No	No
Cluster (date)	Yes	Yes	Yes	Yes
N Panel A (CCGT)	50,824	159,669	50,824	159,669
N Panel A (Hydro)	81,567	132,822	81,567	132,822
N Panel A (H. pump)	17,959	25,207	17,959	25,207
N Panel B (IB)	5,373	45,957	5,373	45,957
N Panel B (GE)	12,507	30,667	12,507	30,667
N Panel B (GN)	28,571	44,464	28,571	44,464
N Panel B (HC)	4,373	38,581	4,373	38,581

Notes: Standard errors in parentheses, clustered by date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Each tech row is a separate regression on that tech’s subsample (the pooled row stacks all three). Unit FE absorbed via within-unit demeaning. Sample: Big-4 (IB/GE/GN/HC) CCGT, Hydro reservoir, Hydro pump-storage sell offers; in-band $|p_{\text{bid}} - \text{MCP}| \leq 140$ EUR/MWh. Critical $h \in \{5, \dots, 8, 16, \dots, 22\}$; flat $h \in \{1, 2, 3\}$; other hours dropped.

Reading. (Gap. Per-tech heterogeneity is large and economically meaningful (Panel A). (i) ID15 — CCGT responds strongest on both margins; Hydro intermediate; pump-storage tiny on σ_p . (ii) DA15 — the leader flips: Hydro reservoir takes the largest σ_p response (+1.69***), CCGT dominant on N_{eff} (+0.44***); pump-storage NEGATIVE on σ_p . (iii) The pooled estimate hides the heterogeneity (e.g. ID15 N_{eff} pooled

null, CCGT +0.79***). Panel B drills into CCGT: ID15 sees a sharp firm split, with IB the biggest σ_p shifter (+6.46***) and GN the biggest N_{eff} shifter (+1.84***); HC essentially null. DA15 looks even more heterogeneous, but the GN row (+35.0*** on σ_p , +3.13*** on N_{eff}) is driven by an anomalous collapse of GN's flat-hour bidding post-reform (pre/post-flat means 40.7 $\rightarrow$ 7.6), so it should be flagged as a level-shift artefact rather than read as a within-hour MTU15 effect. The cleaner DA15 firm story is IB +3.01***, GE -1.47***, HC -12.96***: firms differentiate sharply in how they exploit (or de-exploit) the new granular margin.)

6.4 Spec B — aggregate clearing-price DiD

	(1)	(2)	(3)	(4)
	MTU15-IDA		MTU15-DA	
	B0: no FE	B1: +date FE	B0: no FE	B1: +date FE
Post $\times$ Critical (θ)	-11.32*** (3.72)	-11.18*** (3.75)	+26.70*** (2.51)	+26.70*** (2.54)
<i>Cell means (raw, EUR/MWh)</i>				
Pre, crit	114.0	—	79.4	—
Post, crit	38.3	—	86.9	—
Pre, flat	84.2	—	85.2	—
Post, flat	19.8	—	66.0	—
Date FE	No	Yes	No	Yes
Cluster (date)	Yes	Yes	Yes	Yes
N	8,652	8,652	6,440	6,440

Notes: Outcome: OMIE clearing price (EUR/MWh). Standard errors in parentheses, clustered by date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. B0 is the naive DiD with no FE; B1 adds date FE (the post main effect is absorbed; the DiD is identified strictly off within-day crit-vs-flat variation across the cutover, absorbing every common date-level shock — seasonal, gas-price, reforzada drift, weekday). Sample: ID15 from `marginalpibc` (3 sessions $\times$ period); DA15 from `marginalpdbc`.

Seasonality. The pre/post arms span large seasonal transitions (ID15 winter $\rightarrow$ early spring; DA15 summer $\rightarrow$ autumn) and the level changes are correspondingly large. Spec B1 absorbs every common date-level shock via date FE; the DiD is essentially unchanged (-11.32 $\rightarrow$ -11.18 ID15; +26.70 $\rightarrow$ +26.70 DA15), so the headline numbers are *not* driven by simple date-level seasonality. (*Residual concern, to flag in the prose: a differential seasonal trend in the (crit - flat) differential itself — which date FE does not catch — would still confound the DiD. The standard same-calendar-month robustness against prior years is not directly feasible in the tight windows used here; the cleanest available fallback is the Spec C cross-sectional within-hour dispersion test below, which is post-only and so immune by construction to pre/post seasonal drift.*)

Reading. (Gap. The headline empirical fact: opposite signs. ID15 — critical-hour IDA prices fell EUR11/MWh more than flat-hour ones after MTU15-IDA (granular intraday absorbed within-hour scarcity, compressing the critical peak). DA15 — critical-hour DA prices rose EUR27/MWh more than flat (granular day-ahead pricing extracted higher rents in scarce quarters; within-hour ladder now realised in the clearing price).)

Quantity outcome (DA cleared MW, DA15 only). Same B0/B1 spec but with the OMIE day-ahead cleared MW as outcome (sum across units of `assigned_power_mw` from `pdbc_all`, per (date, period)).

	(1) B0: no FE	(2) B1: +date FE
Post $\times$ Critical (θ , MW)	-2,115*** (255)	-2,115*** (259)
<i>Cell means (raw, MW)</i>		
Pre, crit	20,516	—
Post, crit	19,214	—
Pre, flat	15,643	—
Post, flat	16,457	—
Date FE	No	Yes
Cluster (date)	Yes	Yes
N	6,440	6,440

Notes: Outcome: total DA cleared MW per period. Critical-hour cleared volume *fell* by $-1,301$ MW post-reform; flat-hour rose by $+814$ MW; DiD = $-2,115^{***}$. (Read jointly with the price DiD above: critical-hour DA price ROSE by EUR 27/MWh, cleared QUANTITY FELL by 2,115 MW — the joint pattern is a textbook withholding signature in the critical hours where MTU15-DA created the discrimination margin.)

Reading (quantity). (Gap. The joint price-up / quantity-down pattern in critical hours is the empirical face of strategic discrimination. The same MTU15-DA reform that let firms post higher prices in scarce quarters also let them clear less volume there — consistent with a more elastic supply schedule being pivoted up in critical hours.)

6.5 Spec C — within-hour dispersion (post-MTU15 only)

	(1) D_{price} (unit, EUR/MWh)	(2) D_{price} (unit, EUR/MWh)	(3) D_{qty} (unit, CV)	(4) D_{qty} (unit, CV)	(5) SD_{price} (system, EUR/MWh)	(6) SD_{price} (system, EUR/MWh)
	MTU15-IDA	MTU15-DA	MTU15-IDA	MTU15-DA	MTU15-IDA	MTU15-DA
Critical (θ)	0.39*** (0.05)	1.35*** (0.11)	0.024*** (0.003)	0.033*** (0.002)	4.55*** (0.40)	3.97*** (0.37)
<i>Raw means (post-MTU15 only)</i>						
Mean (flat)	0.12	0.21	0.007	0.008	2.50	2.49
Mean (crit)	0.50	1.53	0.031	0.041	7.04	6.46
Ratio crit/flat	$4.1\times$	$7.3\times$	$4.3\times$	$4.9\times$	$2.8\times$	$2.6\times$
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Unit FE	No	No	No	No	n/a	n/a
Cluster (date)	Yes	Yes	Yes	Yes	Yes	Yes
N	23,797	62,379	23,797	62,379	1,386	1,288

Notes: Cross-sectional regression on the post-MTU15 sample only (pre-MTU15 has one period per hour, so the within-hour SD is undefined). Standard errors in parentheses, clustered by date. Date FE absorbed via within-date demeaning of the outcome and the Critical indicator. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$. Per-unit cells require all 4 within-hour quarters bid; system cells require all 4 quarter clearing prices observed. ID15 post-window: 2025-03-19 to 2025-04-27. DA15 post-window: 2025-10-01 to 2025-12-31.

Reading. (Gap. The falsification: every within-hour dispersion outcome is $3\text{--}7\times$ larger in critical than flat hours post-MTU15. This is the direct confirmation that MTU15 unlocks within-hour discrimination only where within-hour residual demand actually varies. The system-level $SD_{\text{price}} \sim \text{EUR } 6\text{--}7/\text{MWh}$ in critical hours vs $\sim \text{EUR } 2.5$ in flat is the realised cross-quarter clearing-price spread — the empirical face of the unlocked discrimination margin.)

6.6 Falsification — midday hours as a placebo group

If MTU15 only matters where within-hour residual demand actually varies, the DiD should be small on midday hours (11–14, solar-marginal, $\sigma_{\text{within}} \in [468, 502]$ MW — between critical [522, 1303] and flat [336, 357]). Re-running Spec A with midday as the treated group and flat as the control:

	(1) Price spread σ_p	(2)	(3) Tranche count N_{eff}	(4)
	MTU15-IDA	MTU15-DA	MTU15-IDA	MTU15-DA
<i>Post $\times$ Midday (θ), Spec A (Critical replaced by Midday):</i>				
(Pooled)	1.10*** (0.41)	2.26*** (0.43)	0.06 (0.10)	0.18*** (0.04)
CCGT	2.75*** (0.72)	1.55* (0.81)	1.44*** (0.27)	0.40*** (0.06)
Hydro	0.00 (0.47)	2.77*** (0.44)	−0.09 (0.08)	0.05 (0.05)
Hydro pump	0.62 (0.46)	0.38 (0.35)	0.14 (0.10)	0.14** (0.06)
<i>N</i> (pooled)	83,887	129,100	83,887	129,100

Notes: Same Spec A regression as Table 6.3 Panel A but with treated $h \in \{11, 12, 13, 14\}$ instead of $\{5, \dots, 8, 16, \dots, 22\}$. Standard errors in parentheses, clustered by date. Source: `results/regressions/bid/mtu15_critical_flat/specA_falsification_midday.csv`.

Reading. (Gap. The falsification is partial, not clean: midday is not zero. Pooled σ_p midday +1.10*** (ID15) is $\sim 65\%$ of the critical +1.71***; for DA15 the midday +2.26*** actually exceeds the critical +1.36*** pooled. By tech, CCGT midday ID15 (+2.75***) is bigger than CCGT critical ID15 (+2.45***). The honest reading: MTU15 unlocks discrimination wherever within-hour residual demand has any variance — and midday has intermediate variance, so a non-trivial effect there is consistent with the mechanism rather than refuting it. The result is therefore not a pure placebo failure; it sharpens the interpretation as a dose-response reading of MTU15 (effect grows with the within-hour variance of the hour class). A cleaner placebo (e.g. a calendar block with no MTU15 in either arm, or weekends if the variance pattern differs) is the natural next robustness.)

6.7 Further robustness (to add)

(Gap to write. Planned: extended pre windows (full pre-MTU15-IDA back to 2024-06-14; full pre-MTU15-DA back to 2024-06-14) for sample-length sensitivity; same-calendar-month restriction; CCGT scarcity-tier decomposition of the bid-shape outcomes; for DA15, acknowledgement that no clean placebo exists for separating MTU15-DA from reforzada beyond the critical-flat differencing.)